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Mr Ma'am

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SEPM Assignment-

Q.1 Explain Software Process Models.

→ 1. A Software process model is an abstract representation of a software development process. It defines the specific flow of activities actions and tasks required to build high-quality software. The primary goal is to provide structure to what would otherwise be a chaotic development environment, environment ensuring that the final product meets requirements, stays within budgets and its delivered on time

2. Fundamental process Activities

Every model, regardless of its structure, must address these four fundamental activities:

1. Software specification: Defining the functionality and constraints of the software.
2. Software Development: The actual design and programming of the system
3. Software Validation: Ensuring the software does what the customer wants.
4. Software Evolution: Modifying the software to adapt to changing user needs.

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Waterfall Model

waterfall model is a famous and good version of SDLC (System Development life cycle) for software engineering. The waterfall model is a linear and sequential model, which means that a development phase cannot begin until the previous phase is completed. We cannot overlap phases in waterfall model.

Key characteristics: Each phase must be completed and 'signed off' before the next begins.

1. Requirement Analysis: Gathering all possible requirements of the system
2. System Design: Defining the overall system architecture
3. Implementation: coding the system in small units.
4. Integration & testing: combining units and testing the whole system
5. Deployment & maintenance: Releasing the product and fixing post-release issues

Advantages: Simple to Understand, easy to manage due to the rigidity of the model

Disadvantages: Very difficult to go back and change requirements, no working software is produced until late in the life cycle.

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Project Planning

Requirement

Analysis

Design

coding

Testing

Deployment

Incremental Model

~~Spiral Model~~

The ~~Spiral~~ Incremental model is an evolutionary model that each "build" or increment must pass through these specific stages, it combines the elements of the linear model with iterative philosophy.

1. Requirement Analysis: The overall system requirements are defined then specific features are assigned to a particular increment.
2. Design: The architecture of that specific model is planned to ensure it fits with the existing core.
3. Development: The actual programming of the module takes place.
4. Testing: The new increment is tested individually & then integrated with previous versions to check for compatability.

... in a specific state ... time. For eg. the "design" ... might be in the under development state while the "requirements" activity is in the awaiting review state.

Deployment : The functional part is delivered to the customer for use and evaluation

Advantages
users get working software quickly

Disadvantages
high planning - It requires a clear "big picture" before starting

Requirements → Design & Development → testing → implementation
Requirements → Design & Development → testing → Implementation
Requirements → Design & Development → testing → Implementation

Spiral model:

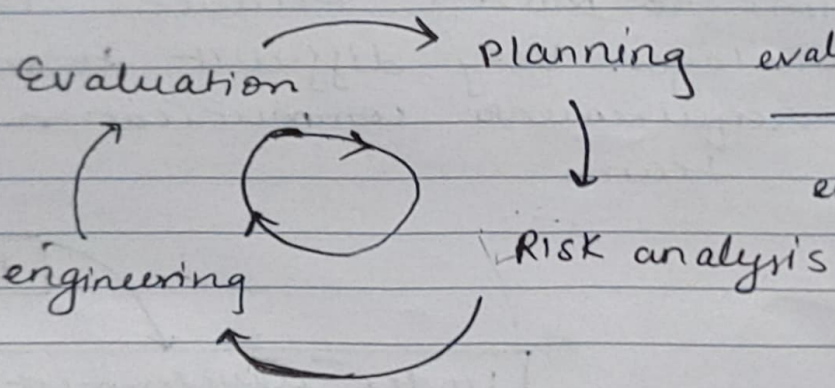
The spiral model is an evolutionary model that ~~each~~ ~~build~~ to combines the iterative nature of prototyping with the controlled aspects of the waterfall model

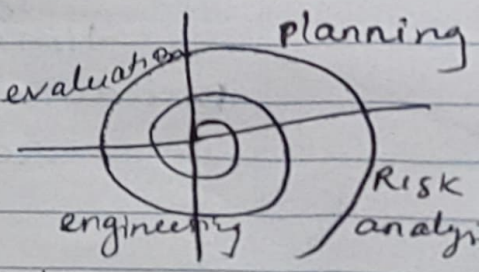
Key characteristics : It is divided into 4 quadrants, The team starts in the center and moves in loops. Each loops involves:

Planning: determining objectives and constraints
Risk analysis: identifying and resolving technical and management risks.
Engineering: developing the next level of the product
Evaluation: customer reviews the progress before the next spiral

Advantages: Strongest model for risk management, suitable for large complex systems.

Disadvantages: can be very expensive, requires high expertics in risk assessment





Concurrent Model

The model is viewed as a nature of activities. Each activity exists in a specific state at any given time. For eg. the "design" activity might be in the under development state, while the "requirements" activity is in the awaiting revision state.

In this model, phases are treated as concurrent activities that communicate with each other.

Communication : Gathering requirements from the client.

Analysis : Modeling the data and functional requirements.

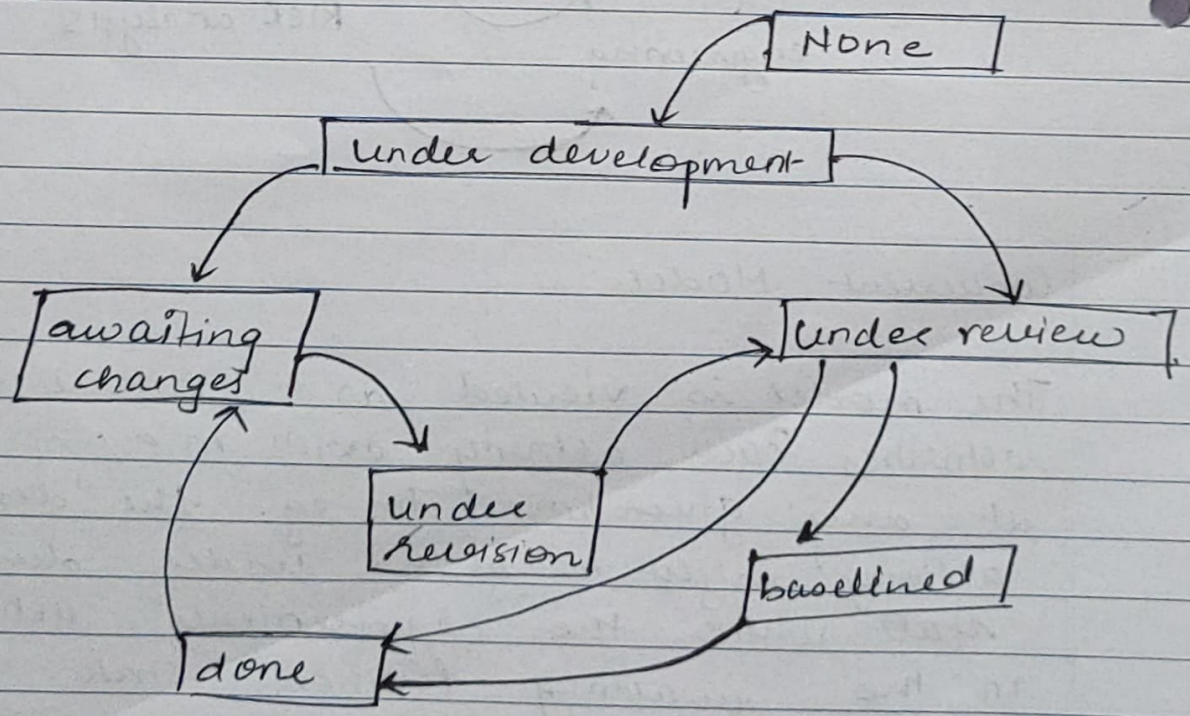
Design : Creating the technical architecture

evolution: coding and building the system

validation: continuous testing as components are completed

Advantages: Significantly reduces the total time to market because tasks overlap.

Disadvantages: Very difficult to manage; requiremens communication between teams



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